

Uncertainty-Aware Out-of-Field Dose Prediction in External Beam Radiotherapy

Mohammed El Aichi^{1,2,4}, Nathan Benzazon^{1,2}, Meïssane M'Hamdi^{1,2}, Rodrigue Allodji³, Florent de Vathaire³, Eric Deutsch^{1,2}, Ibrahima Diallo^{1,2}, Maria Vakalopoulou⁴, Charlotte Robert^{1,2}

¹U1355, Radiothérapie moléculaire et innovations thérapeutiques, Université Paris-Saclay, Gustave Roussy, INSERM, Villejuif, France

²Département de Radiothérapie, Gustave Roussy, Villejuif, France

³U1018, Épidémiologie des radiations, INSERM, Gustave Roussy, Université Paris-Saclay, Villejuif, France

⁴U1361, Cancer Data Science, Université Paris-Saclay, Gustave Roussy, CentraleSupélec, INSERM, Villejuif, France

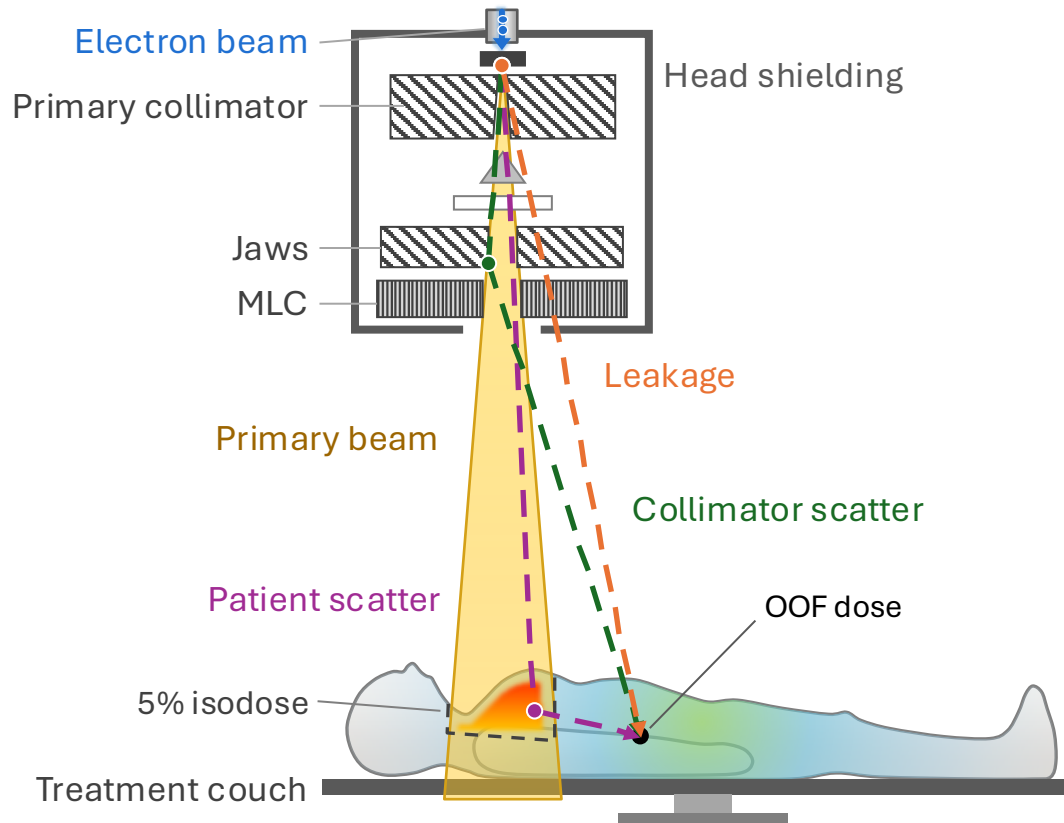
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Oral Session 3: Adaptive Radiotherapy (Longitudinal Imaging, Motion & Dose)



Why out-of-field (OOF) dose matters



OOF dose: low dose beyond the treatment field from patient scatter, collimator scatter, and head leakage.

OOF dose is small, but not zero.

By convention, body voxels receiving less than 5% of the prescribed dose can be considered OOF (e.g. < 3 Gy for a 60 Gy prescription).

OOF dose is linked to long-term health effects.

Including secondary cancers, cardiac toxicity, and cataracts [1].

OOF dose is poorly estimated by treatment planning systems.

Usually inaccurate beyond a few cm from the field edge [1, 2].

[1] Kry et al., AAPM TG-158, Med. Phys. 44(10), 2017.

[2] Howell et al., Phys. Med. Biol. 55, 2010.

How Out-of-field is estimated ?

[1] Jacquet et al., Phys. Med. Biol. 70(15), 2025.
[2] Sánchez-Nieto et al., Front. Oncol. 12, 2022.

Monte Carlo

Jacquet et al. [1]

STRENGTH

Explicit photon transport through the LINAC head and patient.

LIMITATION

Needs proprietary head geometry.

Takes hours per patient on HPC, even with variance reduction [1].

Analytical models

e.g. Periphocal 3D [2]

STRENGTH

Fast whole-body estimates, adequate for cohort studies once calibrated.

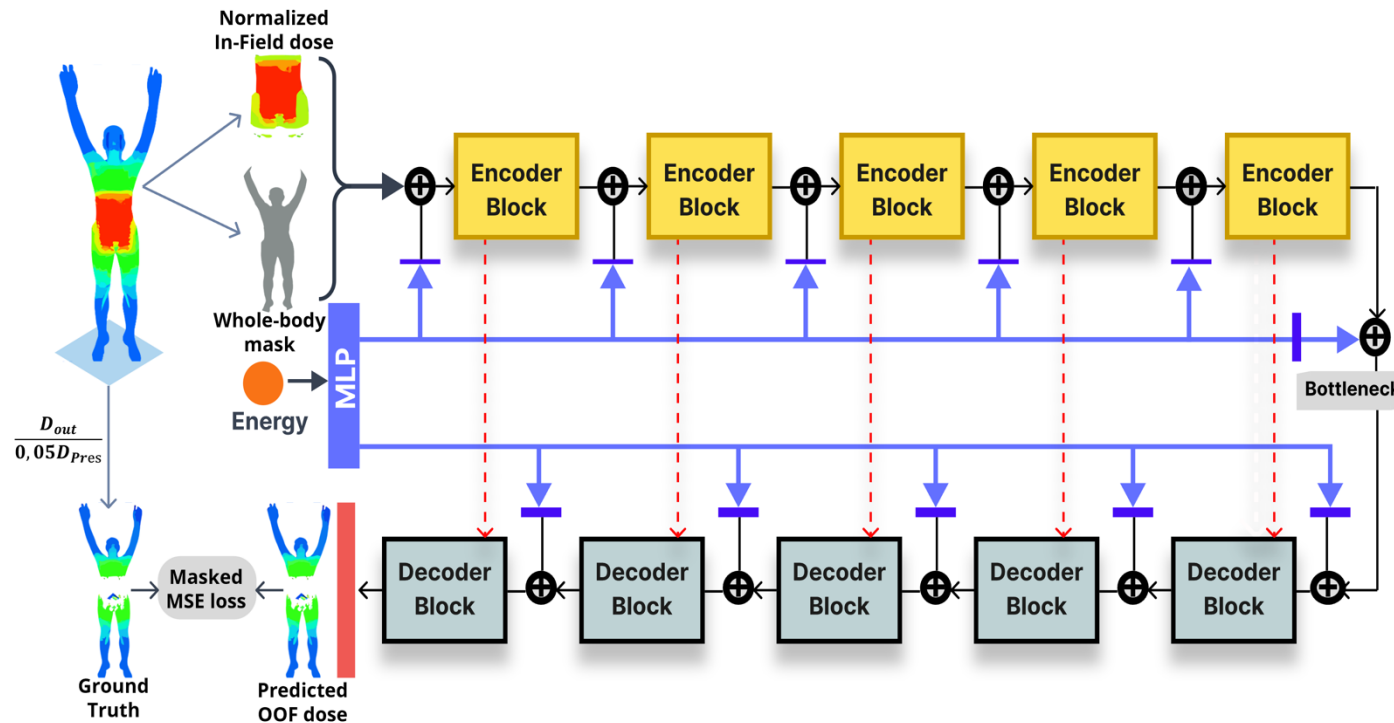
LIMITATION

Requires measurements on each treatment machine, often inaccessible in retrospective cohorts.

If OOF dose can be learned from the planned in-field dose, the whole-body mask, and other information retained in the treatment records, like the beam energy, then we can rapidly reconstruct the OOF dose for large retrospective patient cohorts.

Proposed method (1/2) — Energy-conditioned OOF dose prediction

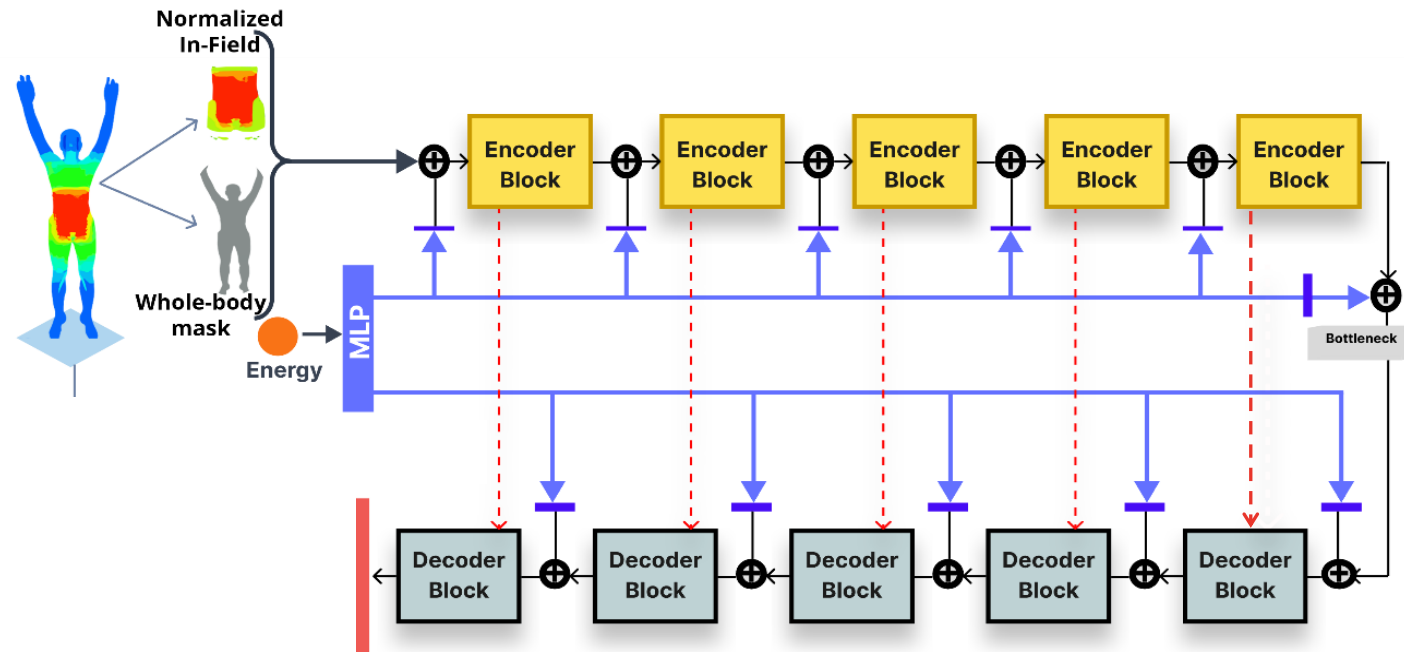
3D voxel-wise regression: normalized in-field dose + binary whole-body mask + beam energy $\rightarrow$ normalized OOF dose.



- 1 Encoder-decoder network**
3D U-Net or Swin UNETR,
trained with MSE on OOF voxels only.
- 2 Beam-energy conditioning**
Energy embedded by a small MLP
and added at every level.

Proposed method (2/2) — Voxel-wise uncertainty estimation

3D voxel-wise heteroscedastic regression: normalized in-field dose + binary whole-body mask + beam energy → mean and log-variance



3 Second output head

Each OOF voxel gets a mean μ and a log-variance $s = \log \sigma^2$.

4 Gaussian-NLL training

Fine-tune the pretrained dose model with Gaussian NLL over OOF voxels:

$$\mathcal{L}_{NLL} = \frac{1}{|\Omega_{OOF}|} \sum_{v \in \Omega_{OOF}} \left(\frac{1}{2} \exp(-\hat{s}(v)) (\hat{\mu}(v) - \check{D}_{OOF}^{GT}(v))^2 + \frac{1}{2} \hat{s}(v) \right)$$

5 Five-fold ensembling

Final dose: average of five predicted means.
Uncertainty: total ensemble variance.

$$\sigma_{ens}^2 = (1/5) \sum_{k=1}^5 [\exp(s_k) + \mu_k^2] - \bar{\mu}^2$$

Dataset: FCCSS Cohort

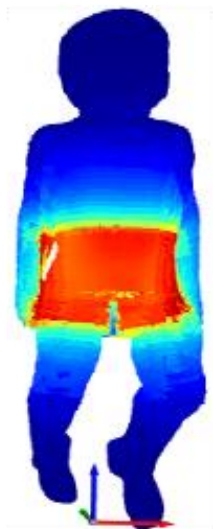


Figure : Example of 3D Dose Distribution Map

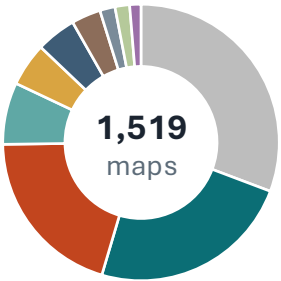
Each record: aligned whole-body dose and body mask, resampled to $4 \times 4 \times 4 \text{ mm}^3$.

Split: 85% train/validation, 15% test; five-fold CV stratified by machine, energy and body volume.

The FCCSS is a French multicenter cohort with 7,205 radiotherapy treatment records from paediatric patients diagnosed with cancer between 1945 and 2000. We selected **1,519 photon LINAC treatments (>1 MV)** for this study.

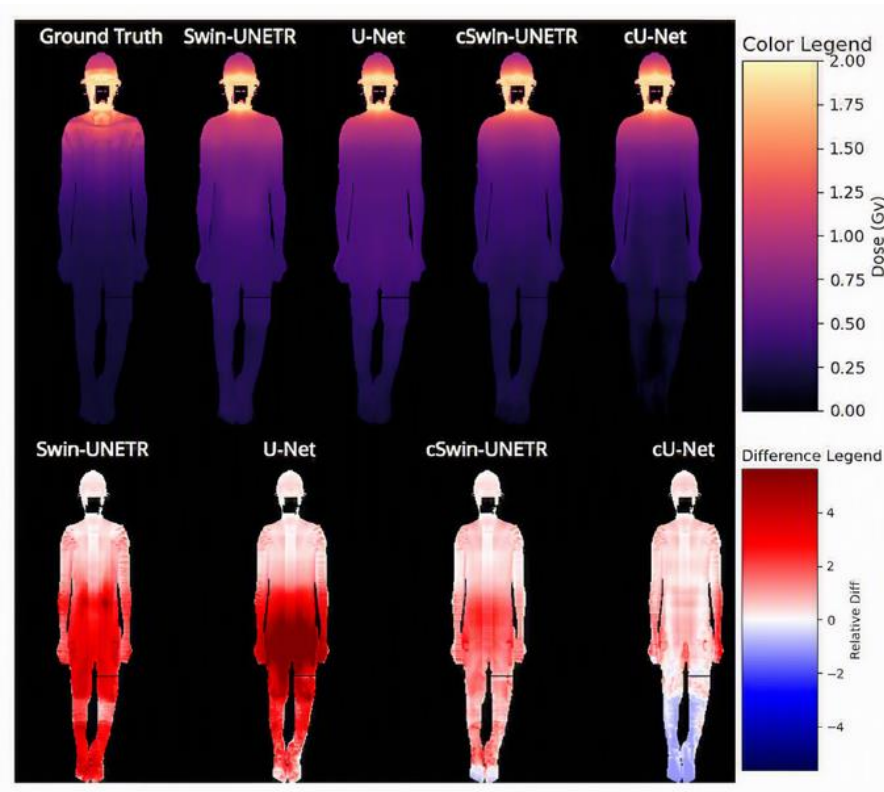
Centre	Maps	Patients	F / M	Age*	Prescription dose*
Institut Curie, Paris	105	91	42 / 49	9.7	40.0
Gustave Roussy, Villejuif	1,345	1,108	563 / 544	10.9	40.0
Institut Godinot, Reims	38	31	26 / 5	8.9	26.0
Claudius-Regaud, Toulouse	31	29	23 / 6	10.7	30.3
Overall	1,519	1,259	654 / 604	10.7	40.0

* Median age (years) and prescription dose (Gy).



- Not available **31%**
- CNS tumors **24%**
- Lymphomas **20%**
- Nephroblastomas **7%**
- Soft-tissue sarcomas **5%**
- Bone tumors **5%**
- Neuroblastomas **3%**
- Other **2%**
- Retinoblastomas **2%**
- Germ cell tumors **1%**

Effect of Beam-Energy Conditioning



Top: predicted OOF dose. Bottom: relative error.

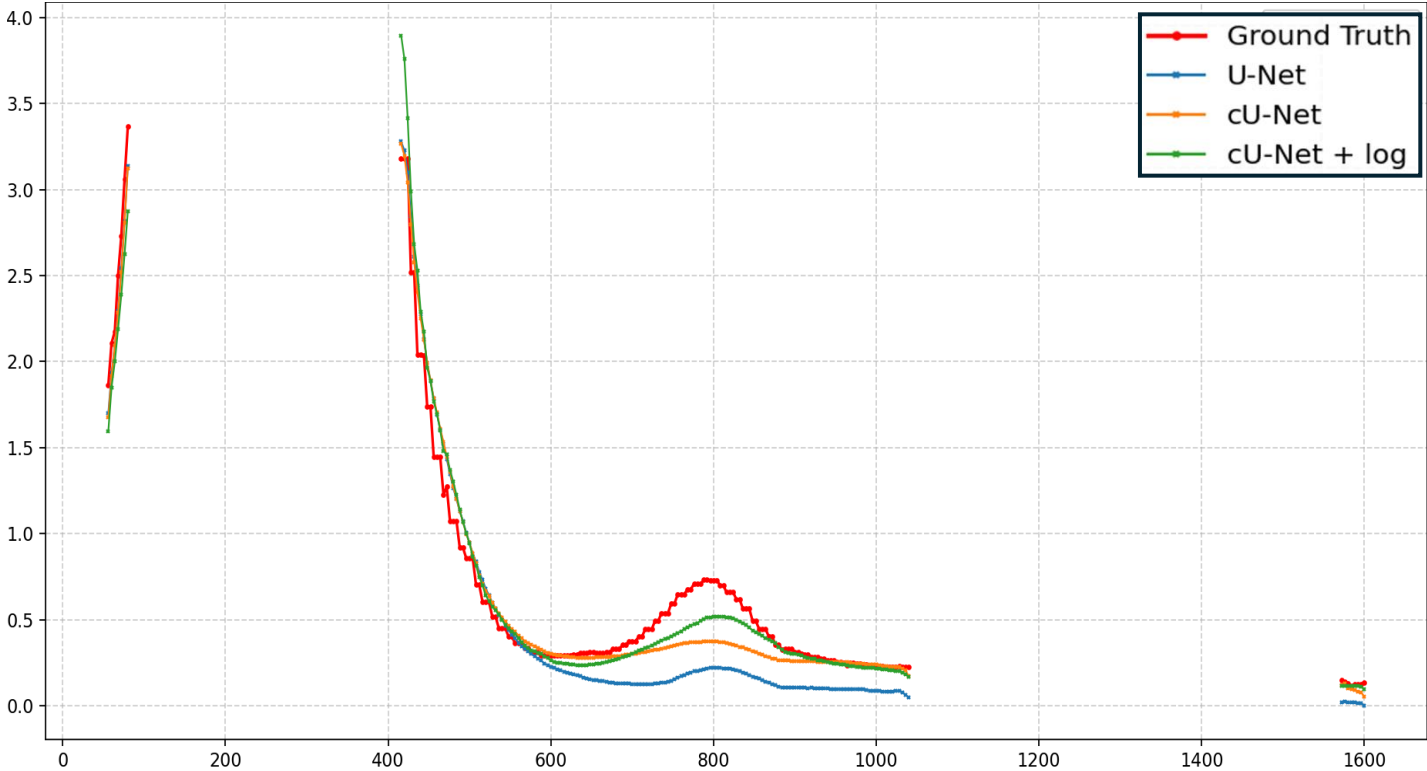
OOF voxels, held-out test · mean (SD)

Model	MAE (cGy)	RMSD (cGy)	PSNR (dB)
U-Net	11.00 (10.01)	16.64 (13.19)	22.53 (4.98)
cU-Net	9.76 (9.28)	15.61 (12.50)	23.13 (5.10)
Swin-UNETR	10.50 (9.74)	15.92 (12.91)	22.94 (5.08)
cSwin-UNETR	10.39 (9.76)	16.22 (12.67)	22.74 (5.09)

U-Net vs cU-Net: MAE 11.00 vs 9.76 cGy ($p < 0.001$)

Paired Wilcoxon signed-rank test.

Target transformation Ablation



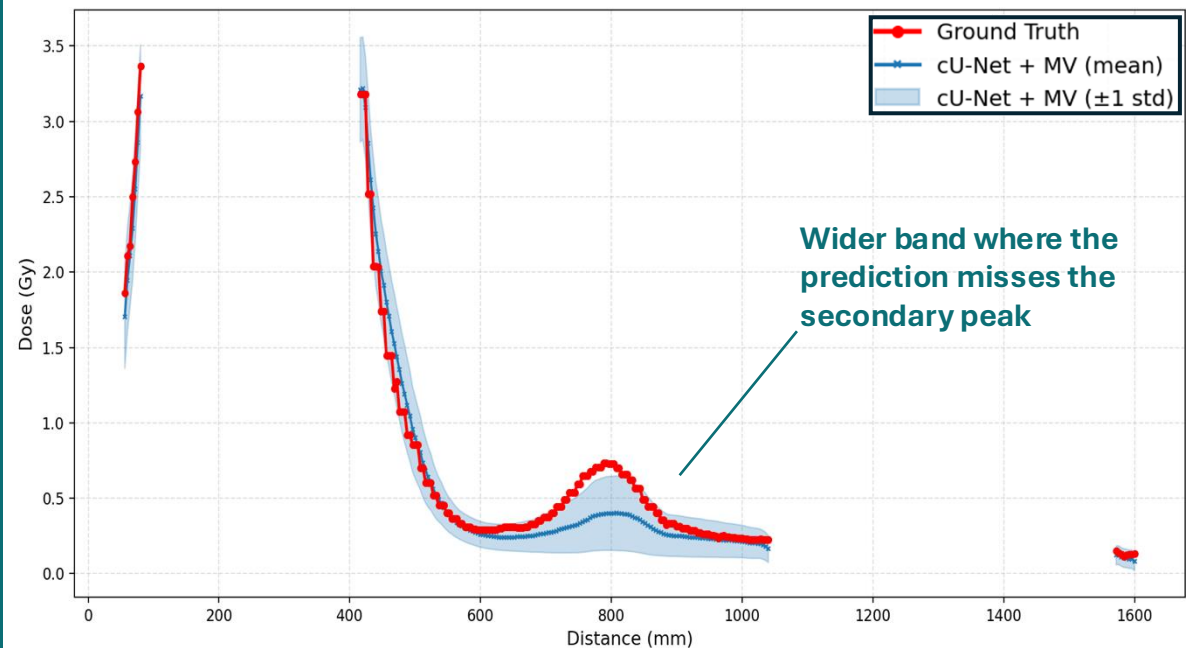
OOF dose profile along the body (Gy vs mm), one test case

Held-out test: mean (SD). REL = relative error by % of prescribed dose

Model	MAE (cGy)	REL (%)		
		1–5%	0.1–1%	0.01–0.1%
cU-Net	9.76 (9.28)	21.1 (7.18)	48.9 (42.10)	258.8 (570.29)
+ sqrt	9.86 (10.15)	22.0 (7.43)	42.6 (38.57)	211.7 (568.79)
+ log	11.32 (12.08)	25.3 (8.71)	43.4 (40.33)	184.7 (497.39)

TARGET TRANSFORMS TRADE PERFORMANCE, log improve REL at 0.01–0.1% (184.68%) but worsen REL at 1–5%.

Mean-variance fine tuning preserves dose accuracy



OOF dose profile · cU-Net + MV mean and ± 1 SD (one test case)

OOF voxels, held-out test · mean (SD)

Model	MAE (cGy)	RMSD (cGy)	PSNR (dB)
cU-Net	9.76 (9.28)	15.61 (12.50)	23.13 (5.10)
cU-Net + MV	9.79 (9.29)	15.70 (12.51)	23.09 (5.13)
cSwin-UNETR	10.39 (9.76)	16.22 (12.67)	22.74 (5.09)
cSwin-UNETR + MV	10.22 (9.71)	16.13 (12.54)	22.81 (5.11)

Similar mean-dose errors after MV fine-tuning, now with voxel-wise uncertainty.

Evaluation of the uncertainty estimation

Per-voxel uncertainty, assessed against prediction error and calibration.

1 Error correlation

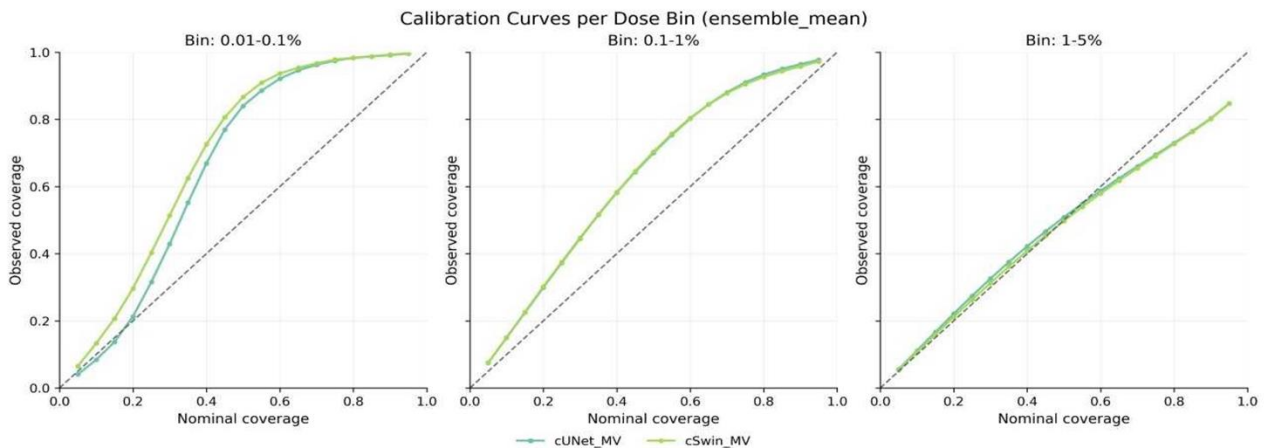
Higher in the lower-dose bins; lower but still positive at 1–5%.

Model	All OOF voxels		0.01–0.1%	0.1–1%	1–5%
	NLL ↓	$\rho_s(e , \sigma) \uparrow$	$\rho_s(e , \sigma) \uparrow$	$\rho_s(e , \sigma) \uparrow$	$\rho_s(e , \sigma) \uparrow$
cU-Net + MV	-1.117	0.594	0.702	0.636	0.391
cSwin-UNETR + MV	-1.069	0.593	0.712	0.657	0.350

2 Calibration

Conservative in the lower-dose bins; closer to the diagonal at 1–5%.

Calibration per dose range (observed vs nominal coverage)



Comparison with prior OOF dose models

Model	Overall		REL (%) by OOF dose range		
	MAE	RMSD	0.01–0.1%	0.1–1%	1–5%
Benzazon et al.	24.28 cGy (10.94)	66.38 cGy (78.98)	1,229% (1538.91)	207.7% (148.86)	51.9% (18.36)
Periphocal 3D	20.16 cGy (12.34)	32.01 cGy (12.05)	430.4% (744.67)	119.1% (151.82)	64.0% (22.66)
cU-Net	7.81 cGy (3.79)	13.17 cGy (5.81)	186.2% (226.87)	46.9% (49.81)	24.2% (5.63)
cU-Net + MV	7.67 cGy (3.56)	13.06 cGy (5.77)	178.6% (208.51)	45.7% (39.68)	24.2% (5.58)

Mean (SD) over the 25 shared cases; REL by OOF dose range (% of prescription).

Conclusion

Energy conditioning helps

MAE 11.0 → 9.8 cGy for U-Net ($p < 0.001$); cU-Net is the best model.

Uncertainty tracks the error

$\rho \approx 0.6$ between predicted σ and $|\text{error}|$, up to 0.7 at the lowest doses, with accuracy preserved.

Beats prior OOF models

7.7 vs 24.3 / 20.2 cGy MAE against Benzazon et al. and Periphocal 3D on 25 shared cases.

Fast, uncertainty-aware whole-body OOF dose the in-field dose, body mask and beam energy.

Code: github.com/maichi98/DoseFieldNet

Limitations & next steps

- Lowest doses remain hard to estimate
- Next: train and validate on Monte Carlo dose; adapt to new machines.

